

IWOTO, Japan

WMO: 479810

Lat: 24.784N

Lon: 141.322E

Elev: 120

StdP: 99.89

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	15.9	16.3	6.2	5.9	17.0	7.2	6.4	17.4	12.0	20.5	11.1	20.4	5.7	20	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.7	31.0	26.3	30.2	26.0	30.1	26.0	27.1	29.2	26.8	28.9	26.5	28.7	6.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.6	22.4	28.5	26.2	21.9	28.3	26.0	21.7	28.2	86.7	29.7	85.5	29.3	84.1	28.2	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.4	10.6	9.5	DB	15.0	31.7	0.8	0.5	14.4	32.0	13.9	32.3	13.5	32.6	12.9	32.9
			WB	10.1	27.6	0.8	0.6	9.5	28.0	9.0	28.4	8.6	28.7	8.0	29.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.4	19.8	19.8	20.7	22.6	24.6	27.0	28.0	27.9	27.8	27.0	25.0	22.0	(6)
(7)		DBStd	3.42	1.49	1.76	1.75	1.72	1.57	1.38	0.93	0.98	0.98	0.99	1.54	1.93	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	11	3	6	2	0	0	0	0	0	0	0	0	1	(9)
(10)		CDD10.0	5249	304	275	333	378	454	511	558	555	533	528	448	371	(10)
(11)		CDD18.3	2218	49	47	76	128	196	261	300	297	283	270	198	113	(11)
(12)		CDH23.3	17523	14	24	72	323	1078	2539	3319	3219	2994	2563	1160	217	(12)
(13)		CDH26.7	3877	0	0	0	7	84	588	968	885	773	475	95	3	(13)
(14)	Wind	WSAvg	5.5	5.7	5.7	5.9	5.7	5.3	4.6	4.9	5.6	5.7	5.5	5.9	5.8	(14)
(15)	Precipitation	PrecAvg	1216	52	39	47	76	156	73	132	207	149	112	106	69	(15)
(16)		PrecMax	1807	107	64	120	199	374	300	337	605	332	392	246	156	(16)
(17)		PrecMin	731	13	6	11	7	26	1	13	32	13	13	23	20	(17)
(18)		PrecStd	272	22	16	28	54	97	59	94	135	87	94	65	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	25.1	26.0	27.2	29.0	31.1	31.2	31.2	31.2	30.2	29.2	27.1	(19)
(20)		0.4%	MCWB	21.6	21.7	22.9	24.1	25.6	26.2	26.2	26.4	26.5	26.2	25.2	23.8	(20)
(21)		2%	DB	23.8	24.1	25.0	26.3	28.1	30.2	31.0	31.0	30.2	30.0	28.2	26.0	(21)
(22)		2%	MCWB	20.5	20.9	22.1	23.4	24.9	25.9	26.2	26.4	26.1	26.0	24.6	23.0	(22)
(23)		5%	DB	23.0	23.1	24.1	25.9	27.8	29.9	30.2	30.2	30.1	29.2	27.9	25.1	(23)
(24)		5%	MCWB	19.6	20.1	21.3	23.2	24.7	25.8	25.9	26.1	26.1	25.6	24.3	22.2	(24)
(25)		10%	DB	22.1	22.2	23.2	25.1	27.0	29.1	29.9	29.9	29.8	28.9	27.1	24.5	(25)
(26)		10%	MCWB	18.5	19.2	20.6	22.8	24.3	25.5	25.8	26.0	25.9	25.4	23.7	21.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.6	22.6	23.6	25.1	26.2	27.3	27.4	27.6	27.3	27.3	26.2	24.7	(27)
(28)		0.4%	MCDWB	23.9	24.0	25.1	26.6	27.9	29.1	29.5	29.4	29.2	28.9	27.9	25.9	(28)
(29)		2%	WB	21.3	21.7	22.7	24.5	25.7	26.7	26.9	27.0	27.0	26.6	25.6	24.0	(29)
(30)		2%	MCDWB	22.9	23.3	24.1	25.8	27.3	28.7	29.3	29.2	29.1	28.5	27.4	25.2	(30)
(31)		5%	WB	20.4	21.0	22.2	23.9	25.3	26.2	26.5	26.6	26.6	26.2	25.1	23.3	(31)
(32)		5%	MCDWB	22.3	22.7	23.6	25.1	26.9	28.3	28.9	28.8	28.8	28.1	27.0	24.7	(32)
(33)		10%	WB	19.4	20.1	21.5	23.4	24.8	26.0	26.2	26.2	26.2	25.9	24.5	22.4	(33)
(34)		10%	MCDWB	21.6	22.1	22.9	24.6	26.3	28.2	28.7	28.5	28.4	27.9	26.5	24.2	(34)
(35)	Mean Daily Temperature Range		MDBR	3.5	3.8	3.8	3.7	3.5	3.6	3.7	3.6	3.5	3.2	3.2	(35)	
(36)		5% DB	MCDBR	4.1	4.3	4.4	4.2	3.9	4.0	4.1	4.0	3.9	3.7	3.5	3.6	(36)
(37)		5% DB	MCWBR	3.5	3.4	3.3	2.7	2.3	2.0	2.0	2.0	2.0	2.1	2.3	2.8	(37)
(38)		5% WB	MCDBR	4.0	4.1	4.1	3.8	3.7	3.7	3.8	3.7	3.7	3.5	3.3	3.5	(38)
(39)		5% WB	MCWBR	4.1	3.7	3.5	2.9	2.5	2.2	2.1	2.1	2.1	2.2	2.4	3.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.379	0.392	0.440	0.461	0.437	0.403	0.417	0.438	0.429	0.418	0.403	0.399	(40)	
(41)		taud	2.472	2.428	2.292	2.282	2.431	2.547	2.504	2.444	2.464	2.488	2.506	2.444	(41)	
(42)		Ebn at Noon	877	890	860	847	861	884	872	856	860	854	846	838	(42)	
(43)		Edh at Noon	99	110	132	136	117	104	108	115	110	103	96	99	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.60	4.46	5.14	5.77	5.94	6.84	6.64	6.09	5.75	4.99	3.94	3.15	(44)	
(45)		RadStd	0.27	0.30	0.37	0.53	0.57	0.51	0.45	0.32	0.34	0.32	0.29	0.32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	+0.32	+0.39	+0.46	N/A	N/A	N/A	N/A	
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page